Design 1

Coarse Mesh:

Max Stress = $1 \cdot 10^2$ ksi
Location: Feature B

Max Displacement: $8.326 \cdot 10^{-2}$ in
Location: Bottom of Feature A

FoS = 0.4

Mass = 0.65 lb

of Elements: 1394

Medium Mesh:

Max Stress = $1.075 \cdot 10^2$ ksi (Location: Feature B)

$$\% = \left| \frac{(1.075 \cdot 10^2) - (1 \cdot 10^2)}{(1.075 \cdot 10^2)} \right| \cdot 100 = \boxed{6.98\%}$$

Converges

Max displacement = $9.555 \cdot 10^{-2}$ in (Location: Bottom Feature A)

$$\% = \left| \frac{(9.55 \cdot 10^{-2}) - (8.326 \cdot 10^{-2})}{(9.55 \cdot 10^{-2})} \right| \cdot 100 = \boxed{12.86\%}$$

Does not converge

FoS = 0.37 $\rightarrow$ $\% = \left| \frac{0.37 - 0.4}{0.37} \right| \cdot 100 = \boxed{7.5\%}$

Does not converge

Mass = 0.65 lb

of Elements: 7807

Design 1 cont.

Fine Mesh

Max Stress = $1.279 \cdot 10^{2}$ ksi (Location: Feature B)

$$\% = \left| \frac{(1.279 \cdot 10^{2}) - (1.075 \cdot 10^{2})}{(1.279 \cdot 10^{2})} \right| \cdot 100$$

$$= \boxed{15.95\%}$$

Does NOT converge

Max displacement = $1.019 \cdot 10^{-1}$ in (Location: Bottom of Feature A)

$$\% = \left| \frac{(1.019 \cdot 10^{-1}) - (9.555 \cdot 10^{-2})}{(1.019 \cdot 10^{-1})} \right| \cdot 100$$

$$= \boxed{6.28\%}$$

CONVERGES

FoS = 0.31 $\rightarrow$ $\% = \left| \frac{0.31 - 0.37}{0.31} \right| \cdot 100 = \boxed{19.35\%}$

Does NOT converge

Mass = 0.651b

of Elements = 65287

Design 2 (Course Mesh)

Max Stress = $6.263 \cdot 10^{1}$ ksi (Location: Top corners of Feature B)

Max displacement = $2.521 \cdot 10^{-2}$ in (Location: Bottom of Feature A)

FoS = 0.64

Mass = 0.661b

Design 2 (cont.)

Medium Mesh

Max Stress = $6.104 \cdot 10^{1}$ ksi (Location: Top corners of Feature B)

$$\% = \left| \frac{(6.104 \cdot 10) - (6.263 \cdot 10)}{(6.104 \cdot 10)} \right| \cdot 100$$

$$= \boxed{2.605\%}$$

Converges

Max Displacement = $2.658 \cdot 10^{-2}$ in (Location: Bottom of Feature A)

$$\% = \left| \frac{(2.658 \cdot 10^{-2}) - (2.521 \cdot 10^{-2})}{(2.658 \cdot 10^{-2})} \right| \cdot 100$$

$$= \boxed{6.213\% \text{ converges}}$$

FoS = 0.66 $\rightarrow$ $\% = \left| \frac{0.66 - 0.64}{0.66} \right| \cdot 100 = \boxed{3.03\%}$

Converges

of Elements: 8390

Mass = 0.6616

Design 3

Course Mesh

Max Stress = 3.261 ksi (Location: Middle of Feature B)

Max Displacement = $2.102 \cdot 10^{-4}$ in (Location: Middle of Feature A)

FoS: 12.27

Mass = 1.4416

of Elements: 1388

Design 3 (cont.)

Medium Mesh

Max Stress = 3.335 ksi (Location: Middle of Feature B)

$\% = \left| \frac{(3.335) - (3.261)}{3.335} \right| \cdot 100 = \boxed{2.22\%}$

Converges

Max Displacement = $2.118 \cdot 10^{-4}$ in (Location: Middle of Feature A)

$\% = \left| \frac{(2.115 \cdot 10^{-4}) - (2.102 \cdot 10^{-4})}{(2.118 \cdot 10^{-4}} \right| \cdot 100$

$= \boxed{0.755\%}$

Converges

FoS = 11.99 → $\% = \left| \frac{11.99 - 12.27}{11.99} \right| \cdot 100 = \boxed{2.34\%}$

Converges

Mass = 1.44 lb

of Elements = 3547

	(Fine Mesh) Design 1	(Medium Mesh) Design 2	(Medium Mesh) Design 3
Max Stress : Location	$1.279 \cdot 10^{2}$ ksi (Feature B)	$6.104 \cdot 10$ ksi (Feature B top corners)	3.335 ksi (Middle of Feature B)
Max deflection : Location	$1.019 \cdot 10^{-1}$ in (Bottom of Feature A)	$2.688 \cdot 10^{-2}$ in (Bottom of Feature A)	$2.118 \cdot 10^{-4}$ in (Middle of Feature A)
Mass	0.651b	0.661b	1.44 lb
Safety Factor (FoS)	0.31	0.66	11.99
# of Elements	65287 (Coarse Mesh : Medium Mesh did NOT converge)	8390 (Both Coarse : Medium converged values)	3547 (Both Coarse : Medium Mesh values converged)

Design 1:

	Course Mesh	Medium Mesh	Fine Mesh
Max Stress & Location	100 ksi, Feature B	107.5 ksi, Feature B	127.9 ksi, Feature B
Max deflection & Location	0.08326 in Bottom of Feature A	0.09555 in Bottom of Feature A	0.1019 in Bottom of Feature A
Mass	0.651 lb	0.651 lb	0.651 lb
FoS	0.4	0.37	0.31
# of Elements	1344	7607	65297

Design 2

	Course Mesh	Medium Mesh
Max Stress & Location	62.63 ksi Top corners of Feature B	61.04 ksi Top corners of Feature B
Max deflection & Location	0.02521 in Bottom of Feature A	0.02688 in Bottom of Feature A
Mass	0.66 lb	0.66 lb
FoS	0.64	0.66
# of Elements	1874	8390

DESIGN 3

	Course Mesh	Medium Mesh
Max Stress & Location	3.261 ksi, Feature B	3.335 ksi, Feature B
Max deflection & Location	0.0002102 in, Feature A	0.0002118 in, Feature A
Mass	1.471 lb	1.47 lb
FoS	12.27	11.99
# of Elements	1388	3547

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